

Jayden Melendez

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EDUCATION

Marist College, Poughkeepsie, New York
B.S. in Computer Science, Minor: Cyber Security

Graduation: May 2026
GPA: 3.95

RELEVANT COURSEWORK

Artificial Intelligence in Software Development, Operating Systems, Algorithms, Computer Organization and Architecture, Intro to z/OS & Major Subsystems, Internetworking, Database Management, Mobile Security

PROFESSIONAL EXPERIENCE

International Business Machine (IBM), Poughkeepsie, New York
Backend Developer Co-op

May 2024 – Present

- **Architectural Modernization:** Contributed to the refactoring of a legacy **50,000+ LoC Java** mainframe patch management system; transitioned tightly-coupled modules into a **Service-Oriented Architecture (SOA)** to improve horizontal scalability and maintainability.
- **Systems Engineering:** Engineered a programmatic diagnostic engine that parses data across hundreds of relational database tables to automate hardware issue detection; reduced manual troubleshooting time by **10%** for 50+ global support.
- **Performance Optimization:** Optimized complex **SQL queries and indexing strategies** within a DB2 environment, achieving a **20% reduction** in existing data retrieval latency for critical patch lifecycle tracking.
- **Reliability & DevSecOps:** Enhanced system stability by implementing a comprehensive **SwaggerUI** testing suite and utilized **Podman** for containerized development, ensuring codebase environmental consistency among team members.
- **Test-Driven Development:** Adopted **TDD** using **JUnit** to ensure the reliability of the refactored SOA modules, reducing production regressions during the mainframe modernization.
- **Collaborative Leadership:** Actively collaborate with senior architects to identify technical debt, utilizing **Agile methodologies** and **Github Enterprise** for code reviews and version control to ensure high-quality software delivery.

Marist College, Poughkeepsie, New York
Resident Assistant

Aug 2023 – Present

- Manage crisis intervention, conflict resolution, and facility logistics for a community of 60+ residents.
- Facilitate community engagement and budget management through coordinated educational programming.

PROJECTS

F.A.S.T — Financial Account Statement Transformer

Feb 2026 – Present

Python, Scikit-learn, Docker, Jenkins, Pydantic, Ollama Phi-4 Mini, YAML

- Engineering a local, containerized **Python DAG pipeline** to automate the extraction and standardization of bank statement data from PDFs into unified CSV schemas using **Ollama's local LLM** to parse the data from the PDFs.
- Implementing a **machine learning router** using **Scikit-learn** (TF-IDF and SGDClassifier) to perform context-aware transaction classification and deduplication across different account types.
- Architecting a robust data validation layer with **Pydantic**, featuring **two-tier reconciliation** (mathematical balance checks and sign validation) to ensure **100% data integrity**.
- Developing a modular, **configuration-driven** system using **Docker** and **YAML**, enabling seamless onboarding of new bank formats without modifying core logic.
- **CI/CD & Quality Assurance:** Implementing a CI/CD pipeline using **Jenkins** to automate unit and module testing suites, ensuring **100% code coverage** for the **Scikit-learn** router and **Pydantic** validation layers before containerized deployment.

NutritionU — A meal planner for students

Aug 2025 – Dec 2025

React, Vite, Node.js, Postgres, Postman

- **Full-Stack Development:** Built a full-stack meal planning application using **React**, **Node.js**, and **PostgreSQL**, enabling students to generate personalized, budget-conscious meal plans based on dietary goals.
- **AI Integration:** Integrated the **OpenAI API** to power intelligent recipe substitution for dietary restrictions, automated macro-nutrient target calculation, and dynamic meal plan generation from user grocery lists.
- **Database Design:** Architected a normalized **PostgreSQL** schema to manage user profiles, grocery inventories, and nutritional data, enforcing data integrity through foreign key constraints and parameterized queries to prevent SQL injection.

SKILLS

Programming Languages: JavaScript, SQL, Python, Java, TypeScript, C++, Assembly, HTML/CSS

Technologies/Frameworks: React, ReactNative, Springboot, Python FastAPI, Node.js, Docker, Linux

Developer Tools: VS Code, GitHub, SwaggerUI, Cursor, Claude Code

EXTRACURRICULARS

Ultimate Frisbee Player, Marist College, Poughkeepsie, NY

Aug 2022 – Present

Spiritual Life and Service Volunteer, Marist College, Poughkeepsie, NY

Aug 2022 – Present